

AGILE SPRINT PLANNING & USER STORY WORKSHOP

Intelligent Public Health Surveillance and Disease Outbreak Prediction System

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1. Project Vision and Stakeholders

1.1 Project Vision

To build an AI-powered, real-time Intelligent Public Health Surveillance and Disease Outbreak Prediction System that integrates healthcare data from hospitals, laboratories, and public health agencies, applies AI and GIS analytics to detect early warning signs of infectious disease outbreaks, and empowers public health authorities to plan resources and respond rapidly — ultimately reducing disease transmission and strengthening community health outcomes.

1.2 Project Objectives

- Collect real-time public health data from hospitals, laboratories, and health agencies.
- Predict disease outbreaks using AI-based analytical models.
- Identify outbreak hotspots through GIS-based spatial mapping.
- Generate early warning alerts for public health officials and field teams.
- Monitor and visualize disease transmission trends over time.
- Support healthcare resource planning and allocation.
- Provide epidemiological analytics dashboards for decision-makers.
- Improve evidence-based public health decision-making.
- Protect sensitive health data through robust access control and compliance measures.
- Enable field-level accessibility even in low-connectivity environments.

1.3 Key Stakeholders

Stakeholder	Role in the Project
Public Health Officials / Ministry of Health	Primary decision-makers who use outbreak predictions and dashboards to plan interventions and policy.
Epidemiologists	Analyze disease trends, validate AI predictions, and interpret outbreak patterns.
Hospital & Clinic Staff	Report patient diagnosis and case data into the surveillance platform.
Laboratory Technicians	Submit confirmed lab test results that feed the disease detection models.
GIS Analysts	Maintain spatial data and hotspot mapping layers.
Data Scientists / AI Engineers	Build, train, and maintain the outbreak prediction models.
IT / Data Engineers	Manage data integration pipelines, cloud infrastructure, and system uptime.
System Administrators	Manage user access, security, and platform maintenance.
Emergency Response Teams	Act on early warning alerts to contain outbreaks in the field.
District Health Administrators	Coordinate local resource allocation based on system recommendations.
NGOs & Community Health Workers	Support outreach and ground-level data collection in underserved areas.
General Public / Citizens	Indirect beneficiaries; may receive public advisories generated by the system.
Compliance & Data Privacy Officer	Ensures the platform meets health-data protection and regulatory requirements.

Stakeholder	Role in the Project
Product Owner	Owns and prioritizes the product backlog based on stakeholder value.
Scrum Master	Facilitates Agile ceremonies, tracks velocity, and removes team impediments.

1.4 Stakeholder RACI Matrix

The matrix below clarifies decision-making authority across the platform's core activities. R = Responsible, A = Accountable, C = Consulted, I = Informed.

Activity	Hosp/Lab Staff	Data Sci Team	GIS Team	Health Officials	Dist. Admins	IT/Compliance
Health Data Collection & Entry	R	A	C	I	I	I
AI Model Development & Validation	I	C	R/A	C	I	I
GIS Mapping & Spatial Analysis	I	C	C	R/A	I	I
Alert Generation & Field Response	I	R	C	I	A	R
Resource Allocation Decisions	I	C	I	I	A	R
Policy & Public Communication	I	C	I	I	R/A	C
System Security & Access Control	I	I	I	I	C	A/R (IT)

1.5 Project Scope

In Scope

- Real-time ingestion of hospital and laboratory case/test data.
- AI-based outbreak risk prediction at the regional level.
- GIS-based hotspot visualization and historical playback.
- Automated multi-channel alerting with escalation.
- Epidemiological trend analytics and executive dashboards.
- Resource planning recommendations (beds, staff, supplies).
- Role-based access control and compliance audit logging.

Out of Scope (Current Phase)

- Direct integration with international (cross-border) health databases.
- Genomic sequencing or pathogen-level laboratory analysis.
- Automated policy enforcement (the system recommends; humans decide).
- Citizen-facing mobile self-reporting application (future phase candidate).

2. Epics and User Stories

The product requirements were broken down into seven Epics, each capturing a major functional area of the system. User stories under each Epic follow the standard Agile format: "As a <user>, I want <feature>, so that <benefit>." Each story also carries its MoSCoW priority, business value, dependency, and estimated story points, established collaboratively during backlog refinement.

EP-01: Real-Time Health Data Collection & Integration

Epic Goal: *Enable hospitals, laboratories, and health agencies to feed real-time case and lab data into the platform.*

Without a reliable, real-time stream of case and lab data, every downstream capability in this platform — prediction, mapping, alerting, and analytics — is built on stale or incomplete information. This epic is treated as foundational precisely because it removes the single biggest source of forecasting error: data latency.

- **US-01 (Must Have, 5 pts):** As a hospital data officer, I want to upload patient diagnosis data in real time, so that public health agencies always have up-to-date disease information for analysis.
- **US-02 (Must Have, 5 pts):** As a laboratory technician, I want to submit confirmed lab test results directly to the surveillance platform, so that confirmed case data is available immediately without manual re-entry or delay.
- **US-11 (Should Have, 5 pts):** As a health agency officer, I want to receive aggregated regional case counts automatically consolidated from multiple hospitals, so that central reporting is standardized and duplicate-free.

EP-02: AI-Based Outbreak Prediction

Epic Goal: Apply AI/ML models to incoming health data to forecast potential outbreaks before they escalate.

Traditional outbreak surveillance relies on manual epidemiological review, which can lag real transmission by days or weeks. Embedding AI directly into the data pipeline compresses that detection window, giving health authorities a meaningful head start on containment before an outbreak becomes a regional crisis.

- **US-03 (Must Have, 13 pts):** As an epidemiologist, I want the system to analyze incoming health data using AI algorithms, so that potential disease outbreaks can be predicted before they escalate.
- **US-04 (Could Have, 8 pts):** As a data scientist, I want to retrain and update prediction models with new historical outbreak data, so that forecast accuracy improves and adapts over time.
- **US-12 (Should Have, 5 pts):** As a data scientist, I want a model performance monitoring dashboard that detects prediction drift, so that the team knows exactly when retraining is required.

EP-03: GIS-Based Hotspot Identification

Epic Goal: *Visualize disease case density spatially to pinpoint geographic outbreak hotspots.*

Disease spread is inherently spatial. A risk score without a map tells an official that something is happening, but a hotspot map tells them where to send resources. GIS visualization turns statistical predictions into an operational, field-actionable picture.

- **US-05 (Must Have, 8 pts):** As a GIS analyst, I want to visualize disease case density on an interactive map, so that outbreak hotspots can be quickly and accurately identified.

- **US-13 (Could Have, 5 pts):** As a GIS analyst, I want a historical hotspot playback feature that replays outbreak spread over time, so that spatial and temporal transmission patterns can be studied.
- **US-14 (Should Have, 5 pts):** As a emergency response team member, I want a mobile-friendly, low-bandwidth version of the hotspot map, so that field teams can access outbreak information even with limited connectivity.

EP-04: Early Warning & Alert System

Epic Goal: Automatically notify relevant stakeholders when outbreak risk thresholds are exceeded.

Even a perfect prediction is worthless if it never reaches the people who can act on it. This epic closes the loop between detection and human response, ensuring alerts reach officials through the channels they actually have access to, including low-connectivity field settings.

- **US-06 (Must Have, 5 pts):** As a public health officer, I want to receive automated early warning alerts when case numbers exceed a defined threshold, so that I can initiate a rapid containment response.
- **US-07 (Should Have, 5 pts):** As a district health administrator, I want alerts to be sent via SMS and email in addition to the dashboard, so that field teams are notified even where internet access is limited.
- **US-15 (Should Have, 3 pts):** As a system administrator, I want unacknowledged alerts to automatically escalate to senior officials after a set time, so that critical outbreak warnings are never missed due to inaction.

EP-05: Disease Trend Monitoring & Epidemiological Analytics

Epic Goal: *Track and visualize disease transmission trends and provide analytics dashboards.*

Point-in-time risk scores are useful, but sustained public-health decision-making requires trend context — is an intervention working, is a region improving or worsening, and how does one district compare to another. This epic gives epidemiologists and policymakers the analytical depth needed for evidence-based decisions.

- **US-08 (Should Have, 5 pts):** As a epidemiologist, I want to view disease transmission trend charts over time, so that I can assess whether current interventions are effective.
- **US-10 (Should Have, 5 pts):** As a health ministry policymaker, I want access to a summarized epidemiological analytics dashboard, so that I can make evidence-based public health decisions quickly.
- **US-16 (Could Have, 3 pts):** As a epidemiologist, I want to compare disease trends across multiple regions side by side, so that relative outbreak severity and spread can be quickly assessed.
- **US-17 (Could Have, 3 pts):** As a health ministry policymaker, I want an automated weekly summary report emailed as a PDF, so that I stay informed without needing to log into the dashboard daily.

EP-06: Healthcare Resource Planning Support

Epic Goal: *Translate outbreak predictions into actionable resource allocation recommendations.*

Prediction and detection only create value if they translate into concrete operational decisions — how many beds, how much staff, how much supply. This epic bridges the gap between forecasting output and healthcare-system readiness.

- **US-09 (Could Have, 8 pts):** As a healthcare resource manager, I want the system to recommend hospital bed, staff, and medical supply allocation based on predicted case loads, so that resources are distributed efficiently ahead of demand.
- **US-18 (Could Have, 3 pts):** As a supply chain officer, I want automated reorder alerts when predicted demand will deplete medical supply stock, so that critical shortages of PPE, test kits, or medication are avoided.

- **US-19 (Could Have, 5 pts):** As a hospital administrator, I want staff shift recommendations based on predicted patient load, so that staffing levels can be optimized ahead of a surge in cases.

EP-07: System Security, Access Control & Compliance

Epic Goal: Protect sensitive health data and ensure the platform meets regulatory and privacy obligations.

A platform handling sensitive patient-level health data must meet strict confidentiality and regulatory obligations from day one. This epic ensures that as the system scales and onboards more hospitals and users, access is tightly controlled and every action is auditable.

- **US-20 (Must Have, 5 pts):** As a system administrator, I want role-based access control for different user types (hospital staff, analysts, officials), so that data confidentiality is maintained and users only access what their role permits.
- **US-21 (Must Have, 3 pts):** As a compliance officer, I want a complete audit log of all data access and modifications, so that the platform can demonstrate compliance with health-data privacy regulations.

3. Acceptance Criteria (Given-When-Then)

Every user story in the backlog has clear, testable acceptance criteria written in the Given-When-Then format, ensuring the development team and QA share a single, unambiguous definition of correct behavior.

For the seven Must-Have stories that form the platform's critical path, a second edge-case scenario is included to demonstrate robustness against missing data, invalid access, or data-correction situations.

Story	Given	When	Then
US-01 (Scenario 1)	the hospital data officer is logged into the surveillance portal with valid credentials	they submit a new patient diagnosis record	the record is validated, timestamped, and stored in the central database within 5 seconds, and an entry is added to the audit log.
US-01 (Scenario 2)	the hospital data officer submits a diagnosis record with a missing required field (e.g., patient region)	the record is validated on submission	the system rejects the submission, highlights the missing field, and does not create a partial record in the central database.
US-02 (Scenario 1)	a laboratory technician has a confirmed test result for a reportable disease	they submit the result through the lab integration API	the case is automatically linked to the corresponding patient record and flagged as "confirmed" in the surveillance system.
US-02 (Scenario 2)	a lab result is submitted for a patient ID that does not yet exist in the system	the linkage service attempts to match the result to a patient record	the system creates a provisional patient record, links the result to it, and flags it for hospital-side confirmation within 24 hours.
US-11 (Scenario 1)	multiple hospitals in a region have submitted case data for the same reporting period	the aggregation job runs at the end of the reporting cycle	the system produces a de-duplicated, consolidated regional case count and flags any conflicting records for manual review.
US-03 (Scenario 1)	sufficient case data exists for a given region over the last 14 days	the AI prediction engine runs its scheduled analysis	the system outputs an outbreak risk score per region with a confidence level, refreshed at least once every 24 hours.

Story	Given	When	Then
US-03 (Scenario 2)	insufficient case data exists for a newly onboarded region (fewer than 14 days of history)	the AI prediction engine attempts to run its scheduled analysis	the system withholds a risk score for that region, displays a "insufficient data" status instead of a numeric prediction, and continues collecting data until the minimum threshold is met.
US-04 (Scenario 1)	new verified outbreak data is available in the historical dataset	a data scientist triggers a model retraining job	the model retrains, is validated against a holdout dataset, and only replaces the production model if accuracy meets or exceeds the current benchmark.
US-12 (Scenario 1)	the production prediction model has been running for a defined evaluation window	the monitoring service compares live prediction accuracy against the training baseline	the system flags the model as "drifted" if accuracy falls below a configured threshold and notifies the data science team.
US-05 (Scenario 1)	case data with valid geolocation exists for a region	a GIS analyst opens the hotspot map view	the map renders a color-coded density layer (low/medium/high risk) that updates as new case data arrives, and supports zoom to district level.
US-05 (Scenario 2)	a region has zero reported cases for the selected period	a GIS analyst views that region on the hotspot map	the region is rendered in a neutral "no data" color distinct from the low-risk color, avoiding a false impression of confirmed safety.
US-13 (Scenario 1)	historical case and risk-score data exists across multiple reporting periods	a GIS analyst selects the "playback" mode and a date range	the map animates hotspot changes chronologically, allowing the analyst to pause, rewind, and step through individual reporting periods.
US-14 (Scenario 1)	a field team member has a mobile device with intermittent connectivity	they open the mobile hotspot map in low-bandwidth mode	the map loads a simplified, low-data version within 5 seconds on a 2G/3G connection and caches the last-synced data for offline viewing.
US-06 (Scenario 1)	the number of new confirmed cases in a region exceeds the configured outbreak threshold	the monitoring engine evaluates the threshold rule	an alert is automatically generated and pushed to the dashboard for all public health officials assigned to that region within 2 minutes.
US-06 (Scenario 2)	case numbers briefly exceed the threshold due to a single data-entry error that is later corrected	the correction is submitted before the alert notification job runs	the system recalculates the count, suppresses the alert, and logs the correction and suppression event for audit purposes.
US-07 (Scenario 1)	an outbreak alert has been generated for a district	the alert notification job runs	an SMS and an email are sent to all registered district health administrators, and delivery status is logged.
US-15 (Scenario 1)	an alert has been sent to a district health administrator and remains unacknowledged after 30 minutes	the escalation checker runs	the alert is automatically escalated to the next-level official and marked as "escalated" in the alert history.
US-08 (Scenario 1)	historical case data exists for a selected region and date range	an epidemiologist selects a region and time period on the trends screen	a line/area chart displays case counts, recovery rate, and transmission rate over the selected period, exportable as PDF or CSV.
US-10 (Scenario 1)	the policymaker has an authenticated account with dashboard access	they log into the analytics dashboard	a summarized view of active outbreaks, hotspot map, trend charts, and resource

Story	Given	When	Then
			status is displayed on a single screen, refreshed in real time.
US-16 (Scenario 1)	two or more regions have available trend data for the same period	an epidemiologist selects them for comparison	the system displays their trend charts side by side on a normalized scale, with key metrics highlighted.
US-17 (Scenario 1)	a policymaker has subscribed to weekly summary reports	the scheduled reporting job runs at the start of each week	a PDF summarizing key metrics, hotspots, and alerts from the prior week is generated and emailed to the subscriber.
US-09 (Scenario 1)	an outbreak risk forecast has been generated for a region	a resource manager opens the resource planning module	the system displays recommended bed capacity, staffing, and supply levels for the next 7-14 days, based on the predicted case load.
US-18 (Scenario 1)	predicted demand for a medical supply item exceeds the current stock level's projected runway	the supply monitoring job evaluates stock against forecast demand	a reorder alert is generated and sent to the supply chain officer with a recommended reorder quantity.
US-19 (Scenario 1)	a predicted patient load has been generated for a hospital	a hospital administrator opens the staffing recommendation view	the system suggests a shift roster adjustment aligned with the predicted patient volume for the next 7 days.
US-20 (Scenario 1)	a new user account is created for the platform	an administrator assigns a role (e.g., hospital staff, analyst, official)	the system restricts that user's access strictly to the features and data permitted for their assigned role.
US-20 (Scenario 2)	a user attempts to access a feature or data view outside their assigned role's permissions	the access-control middleware evaluates the request	the system denies access, returns a clear permission-denied message, and logs the attempted access in the audit trail.
US-21 (Scenario 1)	any user views, creates, or modifies a record in the system	the action is performed	the system logs the user ID, timestamp, action type, and affected record in an immutable audit trail available for compliance review.
US-21 (Scenario 2)	an attempt is made to directly modify or delete an existing audit-log entry	the modification request reaches the audit-log storage layer	the system rejects the modification, since the audit log is append-only, and raises a security alert to the compliance officer.

4. Product Backlog Prioritization

4.1 Prioritized Product Backlog (MoSCoW)

The product backlog was prioritized using the MoSCoW technique, based on business value, dependency on foundational data, and customer/regulatory urgency. Foundational data-ingestion and security stories were ranked highest since nearly every downstream capability depends on them.

Rank	Story	Business Value	MoSCoW	Dependency
1	US-01 - Real-Time Health Data Collection & Integration	High	Must Have	None (foundational)
2	US-02 - Real-Time Health Data Collection & Integration	High	Must Have	None (foundational)

Rank	Story	Business Value	MoSCoW	Dependency
3	US-20 - System Security, Access Control & Compliance	High	Must Have	None (foundational)
4	US-03 - AI-Based Outbreak Prediction	High	Must Have	US-01, US-02
5	US-05 - GIS-Based Hotspot Identification	High	Must Have	US-01, US-02
6	US-06 - Early Warning & Alert System	High	Must Have	US-03
7	US-21 - System Security, Access Control & Compliance	High	Must Have	US-20
8	US-07 - Early Warning & Alert System	Medium	Should Have	US-06
9	US-08 - Disease Trend Monitoring & Epidemiological Analytics	Medium	Should Have	US-01, US-02
10	US-10 - Disease Trend Monitoring & Epidemiological Analytics	Medium	Should Have	US-05, US-08
11	US-11 - Real-Time Health Data Collection & Integration	Medium	Should Have	US-01, US-02
12	US-12 - AI-Based Outbreak Prediction	Medium	Should Have	US-03
13	US-14 - GIS-Based Hotspot Identification	Medium	Should Have	US-05
14	US-15 - Early Warning & Alert System	Medium	Should Have	US-06
15	US-04 - AI-Based Outbreak Prediction	Medium	Could Have	US-03
16	US-09 - Healthcare Resource Planning Support	Medium	Could Have	US-03
17	US-13 - GIS-Based Hotspot Identification	Low	Could Have	US-05
18	US-16 - Disease Trend Monitoring & Epidemiological Analytics	Low	Could Have	US-08
19	US-17 - Disease Trend Monitoring & Epidemiological Analytics	Low	Could Have	US-10
20	US-18 - Healthcare Resource Planning Support	Low	Could Have	US-09
21	US-19 - Healthcare Resource Planning Support	Low	Could Have	US-09

4.2 Team Capacity & Velocity Planning

To size Sprint 1 and Sprint 2 realistically, the team referenced velocity data from prior comparable projects and its own onboarding sprint. A 2-week sprint cadence was adopted with a 6-person cross-functional team (2 backend developers, 1 data scientist, 1 GIS/frontend developer, 1 QA engineer, and 1 Scrum Master / Product Owner pairing).

Reference Sprint	Velocity (Story Points)	Notes
Sprint 0 (Onboarding/Setup)	N/A (setup sprint)	N/A
Prior comparable project - Sprint A	32 points	Baseline reference sprint
Prior comparable project - Sprint B	38 points	Baseline reference sprint
Prior comparable project - Sprint C	35 points	Baseline reference sprint

Reference Sprint	Velocity (Story Points)	Notes
Average Velocity	~35 points / sprint	Used to size Sprint 1 and Sprint 2 commitments

5. Sprint Planning

Based on the prioritized backlog, dependency chains, and team capacity, the team planned two sprints. Sprint 1 establishes the foundational data pipeline and earliest possible outbreak-detection capability. Sprint 2 builds on this foundation by adding security/compliance controls, multi-channel alerting, and analytics dashboards.

5.1 Sprint 1 Backlog & Development Tasks

US-01: As a hospital data officer, I want To upload patient diagnosis data in real time (5 pts)

- Design patient case data schema
- Build secure data-upload API endpoint
- Implement input validation & timestamping
- Build hospital-facing upload UI
- Write unit & integration tests

US-02: As a laboratory technician, I want To submit confirmed lab test results directly to the surveillance platform (5 pts)

- Define lab result data schema
- Build lab integration API (HL7/FHIR compatible)
- Implement patient-record linkage logic
- Add "confirmed case" status flagging
- Write unit & integration tests

US-03: As a epidemiologist, I want The system to analyze incoming health data using AI algorithms (13 pts)

- Prepare and clean training dataset
- Select baseline prediction model (time-series/ML classifier)
- Train and validate initial model
- Build scheduled inference job
- Expose risk-score API for dashboard/GIS consumption

US-05: As a GIS analyst, I want To visualize disease case density on an interactive map (8 pts)

- Set up GIS mapping component (map tile layer)
- Build case-density aggregation service
- Implement color-coded risk layer
- Add district-level zoom & filter controls
- Test with sample geolocation data

US-06: As a public health officer, I want To receive automated early warning alerts when case numbers exceed a defined threshold (5 pts)

- Define configurable threshold rules engine

- Build alert-generation service triggered by risk scores
- Build in-dashboard alert notification UI
- Log alert history
- Write tests for threshold edge cases

Total committed story points: 36.

5.2 Sprint 2 Backlog & Development Tasks

US-20: As a system administrator, I want Role-based access control for different user types (hospital staff, analysts, officials) (5 pts)

- Design role and permission model
- Implement authentication & role-based middleware
- Build admin UI for role assignment
- Restrict API endpoints by role
- Write access-control tests

US-21: As a compliance officer, I want A complete audit log of all data access and modifications (3 pts)

- Design immutable audit-log schema
- Implement logging hooks on create/read/update actions
- Build audit-log viewer for compliance officers
- Add tamper-evidence (hashing/append-only storage)
- Write compliance-focused test cases

US-07: As a district health administrator, I want Alerts to be sent via SMS and email in addition to the dashboard (5 pts)

- Integrate SMS gateway provider
- Integrate email notification service
- Build multi-channel dispatch service
- Implement delivery-status logging
- Test failover between channels

US-08: As a epidemiologist, I want To view disease transmission trend charts over time (5 pts)

- Design trend-chart data aggregation queries
- Build trend visualization component (line/area chart)
- Implement region and date-range filters
- Add PDF/CSV export function
- Write UI and data-accuracy tests

US-10: As a health ministry policymaker, I want Access to a summarized epidemiological analytics dashboard (5 pts)

- Design consolidated dashboard layout
- Integrate hotspot map, trend charts, and resource widgets
- Implement real-time data refresh
- Add role-based dashboard views
- Conduct usability testing with policymaker persona

US-15: As a system administrator, I want Unacknowledged alerts to automatically escalate to senior officials after a set time (3 pts)

- Define escalation timing rules
- Build escalation-checker background job
- Implement escalation notification & logging
- Update alert history UI to show escalation status
- Write tests for escalation edge cases

Total committed story points: 26.

5.3 Future Backlog (Sprint 3+ Candidates)

The remaining backlog items are lower-priority "Should Have" and "Could Have" stories that extend model adaptability, spatial analysis depth, and resource-planning granularity. These are scheduled for future sprints once the core surveillance loop and compliance foundation are stable and validated in production.

- US-04 — Model retraining pipeline (8 pts)
- US-09 — Resource allocation recommendations (8 pts)
- US-11 — Aggregated regional case counts (5 pts)
- US-12 — Model performance monitoring dashboard (5 pts)
- US-13 — Historical hotspot playback (5 pts)
- US-14 — Mobile-friendly low-bandwidth map (5 pts)
- US-16 — Multi-region trend comparison (3 pts)
- US-17 — Automated weekly PDF report (3 pts)
- US-18 — Supply reorder alerts (3 pts)
- US-19 — Staff shift recommendations (5 pts)

6. Story Point Estimation (Fibonacci / Planning Poker)

The team used Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13) to estimate relative effort and complexity for every backlog item. Each member privately selected a card; where estimates diverged significantly, the highest and lowest estimators explained their reasoning before the team re-voted and converged on a consensus value.

Story	Epic	Points	Justification
US-01	Real-Time Health Data Collection & Integration	5	Moderate complexity: standard CRUD/API patterns, a defined integration standard, or a contained UI component.
US-02	Real-Time Health Data Collection & Integration	5	Moderate complexity: standard CRUD/API patterns, a defined integration standard, or a contained UI component.
US-11	Real-Time Health Data Collection & Integration	5	Moderate complexity: standard CRUD/API patterns, a defined integration standard, or a contained UI component.
US-03	AI-Based Outbreak Prediction	13	Very high complexity: requires data preparation, model selection, training, validation, and a new inference pipeline with inherent uncertainty.

Story	Epic	Points	Justification
US-04	AI-Based Outbreak Prediction	8	High complexity: involves a new subsystem (e.g., GIS engine, resource-forecasting logic) with multiple moving parts.
US-12	AI-Based Outbreak Prediction	5	Moderate complexity: standard CRUD/API patterns, a defined integration standard, or a contained UI component.
US-05	GIS-Based Hotspot Identification	8	High complexity: involves a new subsystem (e.g., GIS engine, resource-forecasting logic) with multiple moving parts.
US-13	GIS-Based Hotspot Identification	5	Moderate complexity: standard CRUD/API patterns, a defined integration standard, or a contained UI component.
US-14	GIS-Based Hotspot Identification	5	Moderate complexity: standard CRUD/API patterns, a defined integration standard, or a contained UI component.
US-06	Early Warning & Alert System	5	Moderate complexity: standard CRUD/API patterns, a defined integration standard, or a contained UI component.
US-07	Early Warning & Alert System	5	Moderate complexity: standard CRUD/API patterns, a defined integration standard, or a contained UI component.
US-15	Early Warning & Alert System	3	Small, well-understood scope with minimal integration complexity or a narrow UI/notification addition.
US-08	Disease Trend Monitoring & Epidemiological Analytics	5	Moderate complexity: standard CRUD/API patterns, a defined integration standard, or a contained UI component.
US-10	Disease Trend Monitoring & Epidemiological Analytics	5	Moderate complexity: standard CRUD/API patterns, a defined integration standard, or a contained UI component.
US-16	Disease Trend Monitoring & Epidemiological Analytics	3	Small, well-understood scope with minimal integration complexity or a narrow UI/notification addition.
US-17	Disease Trend Monitoring & Epidemiological Analytics	3	Small, well-understood scope with minimal integration complexity or a narrow UI/notification addition.
US-09	Healthcare Resource Planning Support	8	High complexity: involves a new subsystem (e.g., GIS engine, resource-forecasting logic) with multiple moving parts.
US-18	Healthcare Resource Planning Support	3	Small, well-understood scope with minimal integration complexity or a narrow UI/notification addition.
US-19	Healthcare Resource Planning Support	5	Moderate complexity: standard CRUD/API patterns, a defined integration standard, or a contained UI component.
US-20	System Security, Access Control & Compliance	5	Moderate complexity: standard CRUD/API patterns, a defined integration standard, or a contained UI component.
US-21	System Security, Access Control & Compliance	3	Small, well-understood scope with minimal integration complexity or a narrow UI/notification addition.

Sprint 1 committed effort: 36 story points. Sprint 2 committed effort: 26 story points. Full backlog total estimated effort: 112 story points across 21 user stories.

7. Sprint Goals and Expected Deliverables

7.1 Sprint 1 Goal

By the end of Sprint 1, deliver a working data pipeline that ingests real-time hospital and lab data, generates a baseline AI-driven outbreak risk score per region, visualizes case density on a GIS hotspot map, and

automatically raises an in-dashboard alert when a region crosses its outbreak threshold — establishing the core early-detection loop of the surveillance platform.

Sprint 1 Expected Deliverables

- Functional hospital case-data ingestion API and upload interface.
- Functional laboratory result-submission integration linked to patient records.
- A trained and deployed baseline AI outbreak-prediction model exposing a risk-score API.
- An interactive GIS map showing color-coded disease hotspots by district.
- An automated threshold-based alert mechanism visible on the dashboard.
- Sprint demo showing the end-to-end flow: data upload → prediction → map visualization → alert.
- Updated product backlog and burndown chart at Sprint Review.

7.2 Sprint 2 Goal

By the end of Sprint 2, extend the platform with role-based security and a compliance-ready audit trail, deliver multi-channel (SMS/email) alerting with automatic escalation for unacknowledged warnings, and provide epidemiologists and policymakers with trend-analytics and consolidated dashboard views — moving the platform from a detection prototype toward a field-ready, compliant surveillance system.

Sprint 2 Expected Deliverables

- Role-based access control enforced across all platform features and APIs.
- A tamper-evident audit log covering all data access and modifications.
- Multi-channel (SMS + email) alert delivery with logged delivery status.
- Automatic escalation of unacknowledged alerts to senior officials.
- Disease trend charts with export to PDF/CSV.
- A consolidated executive analytics dashboard combining hotspots, trends, and alert status.
- Updated burndown chart, backlog, and a documented compliance checklist at Sprint Review.

8. Definition of Ready and Definition of Done

8.1 Definition of Ready (DoR)

A backlog item is considered ready to be pulled into a sprint only when it satisfies the following criteria:

- The user story follows the standard "As a / I want / so that" format.
- Acceptance criteria are written in Given-When-Then format and agreed upon by the Product Owner.
- Dependencies on other stories or external systems are identified and either resolved or explicitly scheduled.
- The story has been estimated by the team using Planning Poker.
- Any required design assets, API contracts, or data schemas are available or scoped.
- The story is small enough to be completed within a single sprint.

8.2 Definition of Done (DoD)

A user story is considered complete only when all of the following conditions are met:

- Code is peer-reviewed, merged into the main branch, and passes all automated tests.
- Acceptance criteria for the story are independently verified by QA.
- The feature is deployed to the staging environment and demonstrated at Sprint Review.
- Relevant documentation (API docs, user guides, data schemas) is updated.
- No known critical or high-severity defects remain open against the story.
- Security and access-control implications, if any, have been reviewed and addressed.
- The Product Owner has formally accepted the story.

9. Risk Management and Mitigation Plan

The following risks were identified during sprint planning, along with their likelihood, potential impact, and the team's mitigation strategy.

Risk	Likelihood	Impact	Mitigation Strategy
Incomplete or delayed hospital/lab data feeds	High	High	Build validation and fallback manual-entry workflows; onboard data sources incrementally starting with pilot districts.
AI model produces false positive/negative outbreak alerts	Medium	High	Validate model against historical outbreaks; require epidemiologist sign-off before alerts go live; monitor drift continuously (US-12).
Limited internet connectivity in rural/field locations	High	Medium	Provide SMS-based alerts (US-07) and low-bandwidth mobile map mode (US-14) with offline caching.
Data privacy / regulatory non-compliance	Medium	High	Implement role-based access control (US-20) and full audit logging (US-21) from Sprint 2; conduct a compliance review before go-live.
Alert fatigue among health officials	Medium	Medium	Tune threshold sensitivity with epidemiologist input; introduce escalation logic only for unacknowledged critical alerts (US-15).
Integration complexity with legacy hospital/lab systems	Medium	Medium	Adopt HL7/FHIR standards; run integration spikes early; provide a manual upload fallback UI.
Team unfamiliarity with GIS tooling	Low	Medium	Allocate time for a technical spike on the GIS library; pair less-experienced developers with the GIS analyst.
Scope creep from stakeholder requests mid-sprint	Medium	Medium	Enforce sprint backlog immutability during the sprint; route new requests to the Product Owner for backlog grooming.

10. Agile Ceremonies Schedule

The team follows a standard two-week Scrum cadence, with the following ceremonies structured to keep the backlog healthy, surface blockers early, and maintain stakeholder alignment throughout the project.

Ceremony	Frequency	Duration	Participants	Purpose
Sprint Planning	Start of each sprint (2-week cadence)	2-4 hours	Full Scrum team, Product Owner	Select backlog items, define sprint goal,

Ceremony	Frequency	Duration	Participants	Purpose
				break stories into tasks.
Daily Stand-up	Daily	15 minutes	Development team, Scrum Master	Sync on progress, plans, and blockers.
Backlog Refinement	Mid-sprint, weekly	1 hour	Product Owner, development team leads	Clarify, estimate, and re-prioritize upcoming backlog items.
Sprint Review	End of each sprint	1-2 hours	Full team, Product Owner, key stakeholders	Demo completed increment; gather stakeholder feedback.
Sprint Retrospective	End of each sprint (after review)	1 hour	Development team, Scrum Master	Reflect on process; identify improvements for the next sprint.

11. Sprint Review and Retrospective Framework

11.1 Sprint Review Format

At the end of each sprint, the team demonstrates the working increment to stakeholders, focusing on stories that meet the Definition of Done. Feedback gathered during the review is logged as new or updated backlog items rather than being implemented ad hoc, preserving sprint scope integrity.

11.2 Sprint Retrospective Format

Immediately following the review, the development team holds a retrospective using the "What went well / What didn't go well / Action items" format:

- What went well: Practices, tools, or collaboration patterns the team should continue.
- What didn't go well: Blockers, process gaps, or friction points encountered during the sprint.
- Action items: Concrete, owned, and time-bound changes to try in the next sprint.

Action items from each retrospective are added to the following sprint's backlog as process-improvement tasks and reviewed for completion at the next retrospective, ensuring continuous improvement across the project lifecycle.

12. Glossary of Key Terms

Term	Definition
Epic	A large body of work that can be broken down into a number of smaller, related user stories.
User Story	A short, plain-language description of a feature told from the perspective of the person who wants it, in the format "As a <user>, I want <feature>, so that <benefit>."

Term	Definition
Acceptance Criteria	The specific, testable conditions a user story must satisfy to be considered complete.
Given-When-Then	A structured format for writing acceptance criteria describing the starting context, the triggering action, and the expected outcome.
Product Backlog	The complete, prioritized list of features, fixes, and technical work needed for the product.
Sprint Backlog	The subset of product backlog items selected and committed to for the current sprint.
MoSCoW	A prioritization technique classifying items as Must Have, Should Have, Could Have, or Won't Have.
Story Point	A relative, unit-less measure of the effort and complexity required to implement a user story.
Planning Poker	A consensus-based estimation technique where team members privately select a Fibonacci-sequence card to estimate effort, then discuss discrepancies.
Velocity	The average number of story points a team completes per sprint, used to forecast future capacity.
Sprint Goal	A short statement describing the overarching objective the team aims to achieve by the end of the sprint.
Definition of Ready (DoR)	The criteria a backlog item must meet before it can be pulled into a sprint.
Definition of Done (DoD)	The criteria a piece of work must meet before it can be considered complete.
GIS	Geographic Information System — technology used to capture, analyze, and visualize spatial/geographic data such as disease hotspots.
Outbreak Threshold	A configured case-count or rate-of-change value which, when exceeded, triggers an automated early-warning alert.
RACI Matrix	A responsibility-assignment chart identifying who is Responsible, Accountable, Consulted, and Informed for each activity.

13. Appendix A: Communication Plan

Clear communication channels ensure that information generated by the platform — and information about the platform's development progress — reaches the right people at the right time and through the right medium.

Communication Item	Audience	Frequency	Channel
Daily Stand-up Notes	Development team	Daily	Internal chat channel (async summary)
Sprint Review Demo & Feedback	All stakeholders, Product Owner	End of sprint	Live video call + recorded demo
Outbreak Risk Alerts	Public health officials, district admins, field teams	Real-time (event-driven)	Dashboard + SMS + Email
Weekly Executive Summary	Policymakers, Ministry of Health	Weekly	Automated PDF email report
Backlog Changes / Scope Requests	Product Owner, development leads	As needed	Backlog refinement session
Incident / Escalation Notices	System admins, senior officials	As triggered	Escalation alert + phone call for critical severity

14. Appendix B: Sample Sprint 1 Burndown

The table below illustrates an expected burndown trajectory for Sprint 1's 36 committed story points across a 14-day sprint, based on planned task sequencing and team capacity. Actual burndown is tracked daily on the team's Scrum board and reviewed during stand-ups; any day where remaining points fail to trend downward triggers a mid-sprint impediment discussion.

Sprint Day	Remaining Story Points	Notes
Day 1	36	Sprint planning complete; tasks assigned.
Day 3	31	Data schemas and API scaffolding underway.
Day 5	26	Hospital upload API and lab integration nearing completion.
Day 7	20	Mid-sprint checkpoint; GIS mapping component in progress.
Day 9	13	AI baseline model trained; inference job being wired in.
Day 11	7	Alert engine and dashboard integration in final testing.
Day 13	2	All stories in QA verification; minor bug fixes remaining.
Day 14	0	All Sprint 1 stories meet Definition of Done; ready for review.

A steady, near-linear decline (as opposed to a late-sprint cliff) indicates tasks were broken down to an appropriately granular level during sprint planning and that blockers were surfaced and resolved promptly during daily stand-ups.

15. Appendix C: Team Roles and Responsibilities

The cross-functional team assembled for this project brings together the technical and domain expertise needed to deliver a platform spanning data engineering, machine learning, geospatial visualization, and public-health operations.

Role	Responsibilities	Key Skills
Product Owner	Owns the vision, prioritizes the backlog, and accepts completed stories against acceptance criteria.	Domain knowledge in public health policy; stakeholder management.
Scrum Master	Facilitates ceremonies, tracks velocity/burndown, and removes impediments blocking the team.	Agile facilitation; conflict resolution; servant leadership.
Backend Developers (x2)	Build data-ingestion APIs, alerting services, access-control middleware, and audit logging.	REST API design; database modeling; security best practices.
Data Scientist / AI Engineer	Develops, validates, and monitors the outbreak-prediction and drift-detection models.	Time-series forecasting; ML model validation; Python/ML frameworks.
GIS / Frontend Developer	Builds the hotspot mapping component, dashboards, and mobile-friendly low-bandwidth views.	GIS libraries; data visualization; responsive UI design.

Role	Responsibilities	Key Skills
QA Engineer	Verifies acceptance criteria, writes automated test suites, and validates edge cases before sign-off.	Test automation; exploratory testing; attention to edge cases.
Compliance / Data Privacy Officer	Reviews access-control and audit-logging implementation against regulatory requirements.	Health-data privacy regulation; security auditing.

This structure ensures that every epic in the backlog — from data integration through AI prediction, GIS mapping, alerting, analytics, resource planning, and compliance — has a clear owner capable of driving it from design through to a demoable, production-ready increment.